

# Vol 13 Chapter 2 — Genetic Code and BST Primaries

Keeper (author pass — deep math/physics revision)

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## Chapter 2 — Genetic Code and BST Primaries

*Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine\_DIV5\_QM\_GR\_SM/) or the theorem registry disagree, they win.*

### Level 1 — one sentence

The genetic code — 4 nucleotides (A, T, G, C) encoding 64 codons mapping to 20 amino acids — is a substrate-natural information code whose dimensions  $4 = N_c + 1$ ,  $64 = 4^{N_c}$ , and  $20 = C_2 \cdot N_c + 2$  are BST-primary expressions, with the codon-degeneracy reduction  $64 \rightarrow 20$  as substrate Reed-Solomon compression on  $GF(2^g)$ .

### Level 2 — graduate-physicist precision

#### 2.1 The genetic code

DNA/RNA bases: A (adenine), T or U (thymine/uracil), G (guanine), C (cytosine). 4 bases.

Codons: triplets of nucleotides.  $4^3 = 64$  codons.

Amino acids: 20 canonical (plus 3 stop signals — UAA, UAG, UGA).

The code is nearly universal across all life. Specific codon-amino acid mappings: - Methionine (start): AUG (1 codon) - Leucine: 6 codons (CUU, CUC, CUA, CUG, UUA, UUG) — high degeneracy - Tryptophan, methionine: 1 codon each — low degeneracy

Degeneracy concentrated in 3rd position (wobble).

## 2.2 BST-primary expressions

- 4 nucleotides:  $N_c + 1 = 4$
- 64 codons:  $4^{N_c} = 4^3 = 64$
- 20 amino acids (S-tier structural numerology, demoted per Grace INV-5113 sweep 2026-05-24): Grace's exhaustive 3-primary search found 53 distinct BST-primary expressions equal to 20 from 10 primaries (null-model rate ~5%, well above the Cal #44 risk threshold). Examples:  $C_2 \cdot N_c + 2 = 20$ ,  $N_c + \text{seesaw} = 3+17 = 20$ ,  $g+c_3 = 7+13 = 20$ ,  $\chi+\text{rank}-C_2 = 24-4 = 20$ , ..., plus 48 more. No expression is privileged without a mechanism-forcing derivation. Promotion to I-tier+ requires Lyra task #306 to produce the substrate-dynamics derivation that uniquely picks one expression.
- 23 (20 + 3 stops): cf.  $g \cdot N_c + 2 = 23$ , or M\_24 group order = 24 with one orbit missing
- 3 in "3-nucleotide codon":  $N_c$  literally

## 2.3 Substrate compression view

Codons (64 states) compress to amino acids (20 states) + stops (3 states): factor  $\sim 3$  redundancy.

This is Reed-Solomon-like coding (Vol 14 Ch 9): some redundancy in codon space buys robustness against transcription errors. Wobble base pairing (third position degeneracy) IS the redundancy mechanism.

Substrate view (Paper #122): genetic code is biology's instance of substrate Reed-Solomon coding on  $\text{GF}(2^g) = \text{GF}(128)$ , with codon  $\leftrightarrow$  amino acid as substrate symbol  $\rightarrow$  message-symbol compression.

## 2.4 Universality

Same code across bacteria, archaea, eukaryotes. Minor variants (mitochondrial code, mycoplasma, ciliates) differ in 1-3 codons.

Universality argues for single common ancestor (LUCA) or substrate-forced. BST reads it as substrate-forced: the BST-primary dimensions are the natural code.

## 2.5 K-audit anchors

- Vol 14 Ch 9 (Reed-Solomon on  $\text{GF}(128)$  substrate code)
- Paper #122 (Information substrate)
- K59 cyclotomic (substrate code framework)

### Level 3 — 5th-grader accessibility

Genetic code: DNA has 4 letters (A, T, G, C). Reads in 3-letter words (codons):  $4^3 = 64$ . Codons code for 20 amino acids (plus 3 “stops”). In BST:  $4 = N_c + 1$ ,  $64 = 4^{N_c}$ ,  $20 = S$ -tier structural numerology (53 BST-primary expressions match — no mechanism-forced canonical yet per Grace INV-5113) — first two are substrate-natural, “20” awaits mechanism. Degeneracy (multiple codons  $\rightarrow$  same amino acid) is Reed-Solomon error-correction: the substrate’s universal coding mechanism applied at the genetic scale.

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### What comes next

Chapter 3 develops prebiotic forcing.

### Where to look this up

- Crick, Watson 1953 (DNA structure)
- BST: Vol 14 Ch 9; Paper #122